

Understanding Support Vector Machines

Classification Theoretical Approaches and Applications

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Definition (Binary Classification.)

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the input space and $\mathcal{Y} = \{0, 1\}$ the label set. Binary classification is the task of learning a function

$$h : \mathcal{X} \rightarrow \{0, 1\}$$

from a training set $\{(x_i, y_i)\}_{i=1}^n$, where $x_i \in \mathcal{X}$ and $y_i \in \{0, 1\}$ (or equivalently $\{-1, 1\}$). Many classifiers produce a real-valued score $s(x) \in \mathbb{R}$ and apply a threshold $\tau \in \mathbb{R}$:

$$h(x) = \begin{cases} 1 & \text{if } s(x) \geq \tau, \\ 0 & \text{if } s(x) < \tau. \end{cases}$$

Definition (Linear Classifier).

Let $x \in \mathbb{R}^d$ be an input vector. A *linear classifier* is a function

$$h : \mathbb{R}^d \rightarrow \{-1, +1\}, \quad h(x) = \text{sign}(w^\top x + b),$$

where $w \in \mathbb{R}^d$ is the weight vector and $b \in \mathbb{R}$ is the bias term. The corresponding *decision boundary* is the hyperplane

$$\mathcal{H} = \{x \in \mathbb{R}^d : w^\top x + b = 0\}.$$

More generally, for an output function $f : \mathbb{R} \rightarrow \mathcal{Y}$,

$$h(x) = f(w^\top x + b).$$

Visual Intuition

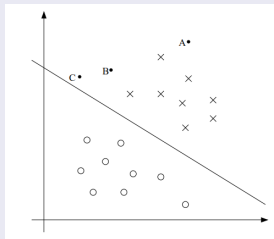


Figure: Linear Classification Example

The figure illustrates a binary classification setting in which positive training examples are denoted by $\times$ and negative training examples by $\circ$. A *separating hyperplane* — the decision boundary defined by the equation $\theta^T x = 0$ — divides the feature space into two half-spaces corresponding to the two classes.

A More Formal Approach on Margins

Definition (Functional Margin).

Given a training example $(x^{(i)}, y^{(i)})$, the *functional margin* of (w, b) with respect to that example is defined as

$$\hat{\gamma}^{(i)} = y^{(i)}(w^T x^{(i)} + b).$$

Given a training set $\mathcal{T} = \{(x^{(i)}, y^{(i)}); i = 1, \dots, m\}$, the functional margin of (w, b) with respect to $\mathcal{T}$ is

$$\hat{\gamma} = \min_{i=1, \dots, m} \hat{\gamma}^{(i)}.$$

Important Property of Functional Margin

Remark.

The classifier $h_{w,b}(x) = g(w^T x + b)$ depends only on the *sign* of $w^T x + b$, not on its magnitude. To see this, note that for any scalar $c > 0$,

$$g(w^T x + b) = g(2w^T x + 2b),$$

so replacing (w, b) with $(2w, 2b)$ leaves $h_{w,b}(x)$ unchanged. However, the functional margin scales linearly with (w, b) :

$$\hat{\gamma}_{(2w, 2b)}^{(i)} = y^{(i)}(2w^T x^{(i)} + 2b) = 2\hat{\gamma}_{(w, b)}^{(i)}.$$

Remark. (cont.)

Hence, by freely rescaling (w, b) , one can make the functional margin $\hat{\gamma}$ arbitrarily large without changing the classifier in any meaningful way. To remove this degeneracy, one may impose the normalization constraint $\|w\| = 1$, i.e., replace (w, b) with

$$\left(\frac{w}{\|w\|}, \frac{b}{\|w\|} \right),$$

and work with the functional margin of the normalized parameters instead.

Geometric Margin (I)

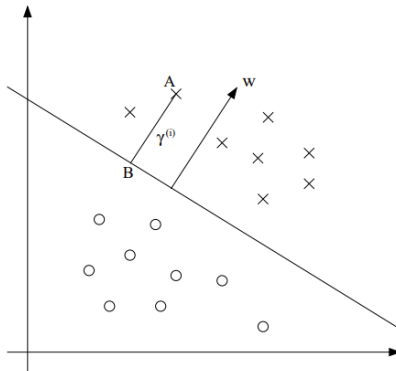


Figure: Geometric Margin Example on Linear Classifier

Geometric Margin (II)

Finding the Geometric Margin

The decision boundary satisfies $w^T x + b = 0$, and w is orthogonal to it. For a positive example $x^{(i)}$ at point A , the projected point B on the boundary is

$$B = x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|}.$$

Since B lies on the boundary, substituting gives

$$w^T \left(x^{(i)} - \gamma^{(i)} \frac{w}{\|w\|} \right) + b = 0.$$

Solving for $\gamma^{(i)}$:

$$\gamma^{(i)} = \frac{w^T x^{(i)} + b}{\|w\|} = \left(\frac{w}{\|w\|} \right)^T x^{(i)} + \frac{b}{\|w\|}.$$

Geometric Margin (III)

Definition (Geometric Margin.)

The *geometric margin* of (w, b) with respect to a training example $(x^{(i)}, y^{(i)})$ is defined as

$$\gamma^{(i)} = y^{(i)} \left(\left(\frac{w}{\|w\|} \right)^T x^{(i)} + \frac{b}{\|w\|} \right).$$

Given a training set $\mathcal{T}$, the geometric margin of (w, b) with respect to $\mathcal{T}$ is

$$\gamma = \min_{i=1, \dots, m} \gamma^{(i)}.$$

Remark.

Note that if $\|w\| = 1$, the functional margin equals the geometric margin, and the geometric margin is invariant to rescaling of (w, b) .

Understanding Max-Margin Classification

Remark.

From discussions before it is precisely correct to conclude that larger geometric margin gives us a more confident classification on our training set.

A decision boundary that maximizes the geometric margin yields a classifier that is not only highly confident in its predictions on the training set, but also achieves a strong fit to the training data — separating the positive and negative examples with the largest possible gap (geometric margin). For now, we will assume that our negative and positive examples can be separated via a hyperplane (decision boundary), that is to say, our data is linearly-separable.

Max-Margin Classification (I)

Main Question

How we can find the hyperplane such that it achieves the maximum geometric margin?

We pose the following optimization problem:

$$\begin{aligned} \max_{\gamma, w, b} \quad & \gamma \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq \gamma, \quad i = 1, \dots, m \\ & \|w\| = 1. \end{aligned}$$

That is, we want to maximize γ , subject to each training example having functional margin at least γ . The $\|w\| = 1$ constraint moreover ensures that the functional margin equals the geometric margin, so we are also guaranteed that all the geometric margins are at least γ . Thus, solving this problem will result in (w, b) with the largest possible geometric margin with respect to the training set.

Max-Margin Classification (II)

Solving the prior optimization problem is enough for finding the max-margin hyperplane which is classifying given data but $\|w\| = 1$ is a non-convex constraint and the form of the optimization problem is hard to implement in some kind of optimization software. Thus we need a "nicer" optimization and constraint on our optimization problem. By using the connection between geometric and functional margins we can transform our problem into:

$$\begin{aligned} \max_{\gamma, w, b} \quad & \frac{\hat{\gamma}}{\|w\|} \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq \hat{\gamma}, \quad i = 1, \dots, m \end{aligned}$$

We got rid of the non-convex constraint but maximizing $\frac{\hat{\gamma}}{\|w\|}$ is again a "nasty" problem to solve.

Max-Margin Classification: Canonical Form

As said before for both margins we can introduce scaling on (w, b) without changing the classifier. We fix the **canonical scaling constraint**:

$$\hat{\gamma} = 1,$$

that is to say, the functional margin of (w, b) with respect to the training set equals 1. Under this normalization, maximizing $\hat{\gamma}/\|w\| = 1/\|w\|$ is equivalent to minimizing $\|w\|^2$. The optimization problem therefore reduces to:

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)} (w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m. \end{aligned}$$

This is a **convex quadratic program** with linear constraints, which can be solved efficiently via standard QP solvers. Mainly, we solved the max-margin problem but we can do it in a better way with using its *dual* which our classification works well on high dimensions and faster than QP.

Primal Optimization Problem

$$\begin{array}{ll}\min_w & f(w) \\ \text{s.t.} & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l.\end{array}$$

Define the **generalized Lagrangian**:

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w),$$

where $\alpha_i \geq 0$ and β_i are the *Lagrange multipliers*.

Primal Objective via $\theta_{\mathcal{P}}$

Define the **primal objective function**:

$$\theta_{\mathcal{P}}(w) = \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta).$$

Key Property

$$\theta_{\mathcal{P}}(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies all primal constraints,} \\ +\infty & \text{otherwise.} \end{cases}$$

Hence, the primal problem is equivalent to:

$$p^* = \min_w \theta_{\mathcal{P}}(w) = \min_w \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{L}(w, \alpha, \beta).$$

Dual Problem

Define the **dual objective function**:

$$\theta_{\mathcal{D}}(\alpha, \beta) = \min_w \mathcal{L}(w, \alpha, \beta).$$

Dual Optimization Problem

$$d^* = \max_{\alpha, \beta : \alpha_i \geq 0} \theta_{\mathcal{D}}(\alpha, \beta) = \max_{\alpha, \beta : \alpha_i \geq 0} \min_w \mathcal{L}(w, \alpha, \beta).$$

Weak Duality

The following always holds using the well-known *min-max inequality*, regardless of convexity:

$$d^* \leq p^*.$$

Under convexity and strict feasibility, we obtain **strong duality**: $d^* = p^*$.

KKT Conditions

Suppose f and g_i are convex, h_i are affine, and the constraints g_i are strictly feasible, that is to say, there exists some w such that $g_i(w) < 0$ for all i . Then w^*, α^*, β^* must satisfy:

Stationarity

$$\frac{\partial \mathcal{L}}{\partial w_i}(w^*, \alpha^*, \beta^*) = 0$$

Dual Feasibility

$$\frac{\partial \mathcal{L}}{\partial \beta_i}(w^*, \alpha^*, \beta^*) = 0$$

Primal Feasibility

$$g_i(w^*) \leq 0, \quad \alpha^* \geq 0$$

Complementary Slackness

$$\alpha_i^* g_i(w^*) = 0$$

Key implication: $\alpha_i^* > 0 \Rightarrow g_i(w^*) = 0$ (the constraint is *active*).

Strong Duality: Summary

Theorem

If f and g_i are convex, h_i are affine, and there exists w such that $g_i(w) < 0$ for all i , then:

$$d^* = p^* = \mathcal{L}(w^*, \alpha^*, \beta^*),$$

and w^, α^*, β^* satisfy the KKT conditions. Conversely, any point satisfying the KKT conditions is a solution to both the primal and dual problems.*

Consequence for SVMs

We can solve the *dual* problem in place of the primal — leading to an algorithm expressed entirely in terms of inner products $\langle x^{(i)}, x^{(j)} \rangle$, which enables the **kernel trick**.

Understanding The Optimization Problem With Lagrangian

As the max-margin classifier we posed an optimization problem:

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)} (w^T x^{(i)} + b) \geq 1, \quad i = 1, \dots, m. \end{aligned}$$

Via using the constraints given in the problem we can write the function $g_i(w)$:

$$g_i(w) = -y^{(i)}(w^T x^{(i)} + b) + 1 \leq 0,$$

as we have one such constraining function g_i for each training sample.

Remark.

From the KKT condition, $\alpha_i > 0$ only for training examples whose functional margin is *exactly* equal to one, i.e., those corresponding to active constraints $g_i(w) = 0$.

Support Vectors (I)

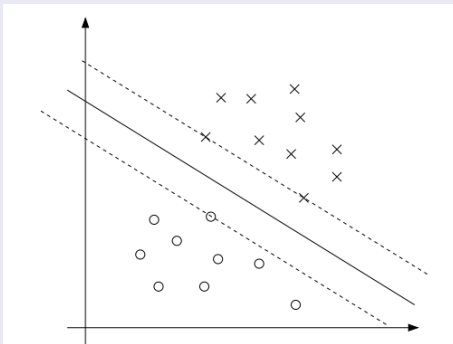


Figure: Max-Margin Classifier with Support Vectors at Dotted Lines

Support Vectors (II)

Definition (Support Vectors).

The **support vectors** are the training examples $(x^{(i)}, y^{(i)})$ for which the corresponding Lagrange multiplier $\alpha_i^* > 0$ at the optimal solution. By the KKT condition,

$$\alpha_i > 0 \implies g_i(w) = 0 \implies y^{(i)}(w^T x^{(i)} + b) = 1,$$

i.e., support vectors are precisely those points lying on the *margin hyperplanes*

$$\mathcal{H}_{\pm} = \{x \in \mathbb{R}^d : w^T x + b = \pm 1\},$$

at geometric distance $\gamma = 1/\|w\|$ from the separating hyperplane $\mathcal{H}_0 = \{x : w^T x + b = 0\}$. All other training examples have $\alpha_i = 0$ and do not contribute to the solution.

Lagrangian of Max-Margin Optimization

We can construct the generalized Lagrangian of the problem as

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i [y^{(i)}(w^T x^{(i)} + b) - 1].$$

For our Lagrangian we only have α Lagrange multiplier since our problem only has one inequality constraint in all dimensions.

Dual of The Problem (I)

To find the dual form, firstly, we need to minimize the Lagrangian with respect to w and b for fixed α to get our dual objective function $\theta_{\mathcal{D}}$. As the method proposed before we will start by setting the derivatives of $\mathcal{L}$ with respect to w and b to zero. We have

$$\nabla_w \mathcal{L}(w, b, \alpha) = w - \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} = 0.$$

This gives us a new form for w :

$$w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$$

Dual of The Problem (II)

With derivative with respect to b , we obtain

$$\frac{\partial}{\partial b} \mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i y^{(i)} = 0.$$

By substituting our new form for w to the Lagrangian we get

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - b \sum_{i=1}^m \alpha_i y^{(i)}.$$

But from last equality we already know the last term is zero

$$\mathcal{L}(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)}.$$

Dual of The Problem (III)

By putting the constraints $\alpha_i \geq 0$ and $\sum_{i=1}^m \alpha_i y^{(i)} = 0$, we get the dual problem:

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle \\ \text{s.t.} \quad & \alpha_i \geq 0, \quad i = 1, \dots, m \\ & \sum_{i=1}^m \alpha_i y^{(i)} = 0. \end{aligned}$$

One can check sufficient conditions for $p^* = d^*$ and the KKT conditions hold.

Recovering the Primal Solution

Once the dual problem is solved (i.e., the α_i 's maximizing $W(\alpha)$ are found), the optimal weight vector is recovered via:

$$w^* = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}.$$

Optimal Intercept b^*

Having found w^* , the optimal bias term is:

$$b^* = - \frac{\max_{i: y^{(i)} = -1} w^{*T} x^{(i)} + \min_{i: y^{(i)} = 1} w^{*T} x^{(i)}}{2}.$$

Key Observation

The dual is a maximization problem in the parameters α_i . Solving it and using the given form for w gives the optimal w^* , from which b^* follows directly by the equation above.

Prediction via Inner Products & The Kernel Trick

After solving the dual, the prediction at a new point x is:

$$w^T x + b = \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x + b = \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b. \quad (12-13)$$

We predict $y = 1$ if and only if $w^T x + b > 0$. Since $\alpha_i = 0$ for all non-support vectors, only inner products with support vectors need to be computed. Thus using this property of the problem we can use something called *kernel trick* and then we can call this classification model algorithm **support vector machine**.

Feature Mappings & Kernels

Rather than applying SVMs on raw input attributes x , we may learn using a **feature mapping** $\phi : \mathbb{R}^d \rightarrow \mathcal{F}$.

Example

For a cubic regression, the feature mapping is:

$$\phi(x) = \begin{bmatrix} x \\ x^2 \\ x^3 \end{bmatrix}.$$

Definition (Kernel Function)

Given a feature mapping ϕ , the corresponding **kernel** is defined as:

$$K(x, z) = \phi(x)^T \phi(z).$$

Replacing $\langle x, z \rangle$ with $K(x, z)$ everywhere in the algorithm allows implicit computation in $\mathcal{F}$ without ever explicitly evaluating $\phi(x)$.

Polynomial Kernel: Example

Let $x, z \in \mathbb{R}^n$ and consider $K(x, z) = (x^T z)^2$:

$$K(x, z) = \left(\sum_{i=1}^n x_i z_i \right)^2 = \sum_{i,j=1}^n (x_i x_j)(z_i z_j) = \phi(x)^T \phi(z),$$

where $\phi(x) = [\dots, x_i x_j, \dots]^T$ contains all degree-2 monomials.

Computational Advantage

- Computing $\phi(x)$ explicitly requires $O(n^2)$ time.
- Computing $K(x, z) = (x^T z)^2$ requires only $O(n)$ time.

More generally, $K(x, z) = (x^T z + c)^d$ corresponds to a feature space of all monomials up to degree d , of dimension $\binom{n+d}{d}$, yet costs only $O(n)$ to evaluate.

Kernels as Similarity Measures

Intuition

If $\phi(x)$ and $\phi(z)$ are close, then $K(x, z) = \phi(x)^T \phi(z)$ is large. If they are nearly orthogonal, $K(x, z) \approx 0$. Hence K measures *similarity* between x and z in feature space.

A natural candidate for a similarity-based kernel is the **Gaussian (RBF) kernel**:

$$K(x, z) = \exp\left(-\frac{\|x - z\|^2}{2\sigma^2}\right).$$

- $K(x, z) \approx 1$ when x and z are close.
- $K(x, z) \approx 0$ when x and z are far apart.
- Corresponds to an **infinite-dimensional** feature mapping ϕ .

Mercer's Theorem: Valid Kernels

Theorem (Mercer)

Let $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Then K is a valid (Mercer) kernel if and only if for any finite set $\{x^{(1)}, \dots, x^{(m)}\}$, the **kernel matrix** $K \in \mathbb{R}^{m \times m}$ defined by

$$K_{ij} = K(x^{(i)}, x^{(j)})$$

is **symmetric positive semi-definite**.

Implication

To verify that a function K is a valid kernel, it suffices to check that the kernel matrix is symmetric and positive-definite — no need to explicitly construct a feature mapping ϕ .

The Kernel Trick: General Principle

Definition (Kernel Trick)

If a learning algorithm can be written entirely in terms of inner products $\langle x, z \rangle$ between input vectors, then replacing each inner product with a kernel $K(x, z)$ allows the algorithm to operate implicitly in the high-dimensional feature space corresponding to K :

$$\langle x, z \rangle \mapsto K(x, z) = \phi(x)^T \phi(z).$$

Prediction rule becomes:

$$w^T x + b = \sum_{i=1}^m \alpha_i y^{(i)} K(x^{(i)}, x) + b.$$

- Applicable beyond SVMs: kernel perceptron, kernel PCA, Gaussian processes, etc.
- Computation stays $O(n)$ even when $\dim(\mathcal{F}) = \infty$.

Regularization & The Non-Separable Case

The hard-margin SVM assumes *linear separability*. In practice:

- Data may not be linearly separable.
- A separating hyperplane may be highly sensitive to outliers.
- Kernel trick generally gives a well-behaving classifier hyperplane but sometimes it is not enough.

Problem

A single outlier can cause the decision boundary to swing drastically, resulting in a much smaller margin and poor generalization.

We relax the hard constraints by introducing **slack variables** $\xi_i \geq 0$, allowing examples to violate the margin by a controlled amount — leading to the **soft-margin SVM**.

ℓ_1 Soft-Margin Formulation

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i, \quad i = 1, \dots, m \\ & \xi_i \geq 0, \quad i = 1, \dots, m. \end{aligned}$$

- $\xi_i > 0$: example i violates the margin by ξ_i .
- $C > 0$: trade-off between margin size and constraint violation.
 - Large C : penalize violations heavily $\rightarrow$ smaller margin.
 - Small C : tolerate violations $\rightarrow$ larger margin.

ℓ_2 Soft-Margin Formulation

$$\begin{aligned} \min_{\gamma, w, b} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i, \quad i = 1, \dots, m \\ & \xi_i \geq 0, \quad i = 1, \dots, m. \end{aligned}$$

- Penalizes the *square* of slack, making the objective strongly convex.
- Produces a **unique** solution and is more sensitive to large violations than ℓ_1 .
- The dual takes the same form as ℓ_1 but with a modified box constraint: $\alpha_i \geq 0$ (no upper bound C).

Dual of the Soft-Margin SVM

The dual problem of the ℓ_1 soft-margin SVM is:

$$\begin{aligned} \max_{\alpha} \quad & W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C, \quad i = 1, \dots, m \\ & \sum_{i=1}^m \alpha_i y^{(i)} = 0. \end{aligned}$$

KKT Dual Complementarity Conditions

$$\begin{aligned} \alpha_i = 0 &\Rightarrow y^{(i)}(w^T x^{(i)} + b) \geq 1, & \alpha_i = C &\Rightarrow y^{(i)}(w^T x^{(i)} + b) \leq 1, \\ 0 < \alpha_i < C &\Rightarrow y^{(i)}(w^T x^{(i)} + b) = 1. \end{aligned}$$

Hard-Margin vs. Soft-Margin SVM

Hard-Margin SVM

$$\begin{aligned} \min_{w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1 \end{aligned}$$

- Requires *linear separability*.
- No tolerance for misclassification.
- Dual: $0 \leq \alpha_i$.
- Sensitive to outliers.

Soft-Margin SVM

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i \\ & \xi_i \geq 0 \end{aligned}$$

- Works on non-separable data.
- Tolerates margin violations via ξ_i .
- Dual: $0 \leq \alpha_i \leq C$.
- Robust to outliers.

In the statistical learning theory framework, an **algorithm** is a strategy for choosing a function $f : X \rightarrow Y$ given a training set $\mathcal{T} = \{(x_1, y_1), \dots, (x_n, y_n)\}$, where $y_i \in \{-1, +1\}$.

Regularized Risk Minimization (General Form)

$$f = \operatorname{argmin}_{f \in \mathcal{H}} \left\{ \frac{1}{n} \sum_{i=1}^n V(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}}^2 \right\},$$

where $\mathcal{H}$ is a *hypothesis space*, $V : Y \times Y \rightarrow \mathbb{R}$ is the loss function, $\|\cdot\|_{\mathcal{H}}$ is a norm on $\mathcal{H}$, and $\lambda \in \mathbb{R}$ is the **regularization parameter**.

- Empirical risk $\frac{1}{n} \sum_i V(y_i, f(x_i))$ measures fit to data.
- Regularization term $\lambda \|f\|_{\mathcal{H}}^2$ penalizes complexity to avoid overfitting.

SVM as Empirical Risk Minimization

The soft-margin SVM fits exactly into this framework with the **hinge loss**:

$$V(y_i, f(x_i)) = \max(0, 1 - y_i f(x_i)).$$

SVM as Regularized ERM

$$\min_{w, b} \quad \frac{1}{n} \sum_{i=1}^n [1 - y_i(w^T x_i + b)]_+ + \lambda \|w\|^2,$$

which is equivalent (with $\lambda = \frac{1}{2C}$) to the soft-margin primal:

$$\min_{w, b, \xi} \quad \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i, \quad \text{s.t. } y_i(w^T x_i + b) \geq 1 - \xi_i, \xi_i \geq 0.$$

Reproducing Kernel Hilbert Space (RKHS)

When $\mathcal{H}$ is a **reproducing kernel Hilbert space (RKHS)**, there exists a kernel $K : X \times X \rightarrow \mathbb{R}$ representable as an $n \times n$ symmetric positive-definite matrix K .

Representer Theorem

The optimal $f \in \mathcal{H}$ admits the representation:

$$f(x_i) = \sum_{j=1}^n c_j K_{ij},$$

$$\|f\|_{\mathcal{H}}^2 = \langle f, f \rangle_{\mathcal{H}} = \sum_{i=1}^n \sum_{j=1}^n c_i c_j K(x_i, x_j) = c^T K c.$$

Consequence

The infinite-dimensional optimization over $f \in \mathcal{H}$ reduces to a finite-dimensional optimization, computable via the kernel matrix K .

SVM in RKHS: Kernel ERM Form

Substituting the representer theorem into the regularized ERM gives the **kernelized SVM objective**:

$$\min_{c \in \mathbb{R}^n} \frac{1}{n} \sum_{i=1}^n \left[1 - y_i \sum_{j=1}^n c_j K(x_j, x_i) \right]_+ + \lambda c^T K c.$$

Linear SVM

$$K(x, z) = x^T z$$

$$f(x) = w^T x + b$$

$$\|f\|_{\mathcal{H}}^2 = \|w\|^2$$

Kernel SVM

$$K(x, z) = \phi(x)^T \phi(z)$$

$$f(x) = \sum_j \alpha_j y_j K(x_j, x)$$

$$\|f\|_{\mathcal{H}}^2 = c^T K c$$

Key Insight

The kernel trick turns the SVM into a *finite-dimensional convex program* regardless of the (possibly infinite) dimension of the feature space $\mathcal{H}$.

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